

The Limits of Predictions for Online Bipartite Matching

A Unified Experimental Study

Zhuolun Li

MSc in Computer Science · School of Computer Science · University of Bristol

Predictions are robustness insurance, not a performance lever.

0.990

the advice-free baseline

0.009

left for perfect advice

0.472

unguarded following

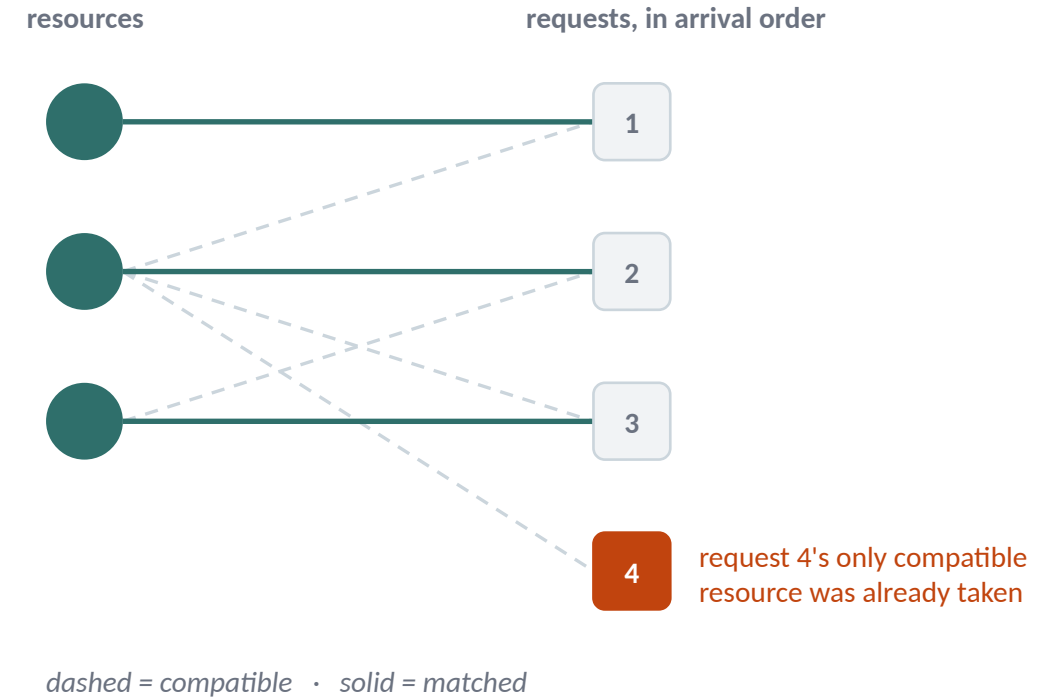
THE PROBLEM

Irrevocable decisions under uncertainty

- Resources are known in advance; requests arrive one at a time.
- Each request must be matched to a free compatible resource, or dropped, before the rest of the input is seen.
- No decision can be undone.

*Online advertising · ride-hailing dispatch · AI-inference
request routing*

Performance = the matching produced ÷ the offline optimum on the same instance.



Two guarantees, two families of algorithm

Consistency

near-optimal when the prediction is good

Robustness

no worse than a prediction-free algorithm when it is wrong

1

MinPredictedDegree

consumes a prediction of how contended each resource will be, and serves the scarcest first

prediction: one number per resource

2

Test-and-fallback

tests a prediction of the arrival mix on a short prefix of the requests, then commits to following it or falling back

prediction: a histogram over request types

Both were analysed only in isolation — each on its own inputs, its own notion of prediction error, and almost entirely in worst-case theory.

Three questions

1

How do the two families actually compare, head to head, under one prediction-error model — and how much does a prediction buy on realistic data?

answered experimentally

2

What are the failure modes of the adaptive test-and-fallback mechanism: the cost of its test, and the calibration of its accept/reject decision?

answered experimentally

3

Why does the average-case experience differ so sharply from the worst-case promise — is the wall an artefact, or is it necessary?

answered in outlook form

The first two are answered experimentally; the third in outlook form, and I say exactly how far that goes.

One harness: only the prediction varies

1

Same instances

identical graphs, arrival sequences, offline optimum (Hopcroft-Karp) and paired random streams for every algorithm

2

Structured error

four error models injected along the structure of the instance, not as i.i.d. noise

3

Stated uncertainty

paired trials, 95% confidence intervals on every reported ratio

4

Reproducible

every figure and table regenerated from a fixed seed by one script

Validated before use

I reproduced the published experimental study of Borodin, Karavasilis and Pankratov before adding any prediction-based work.

5 / 5

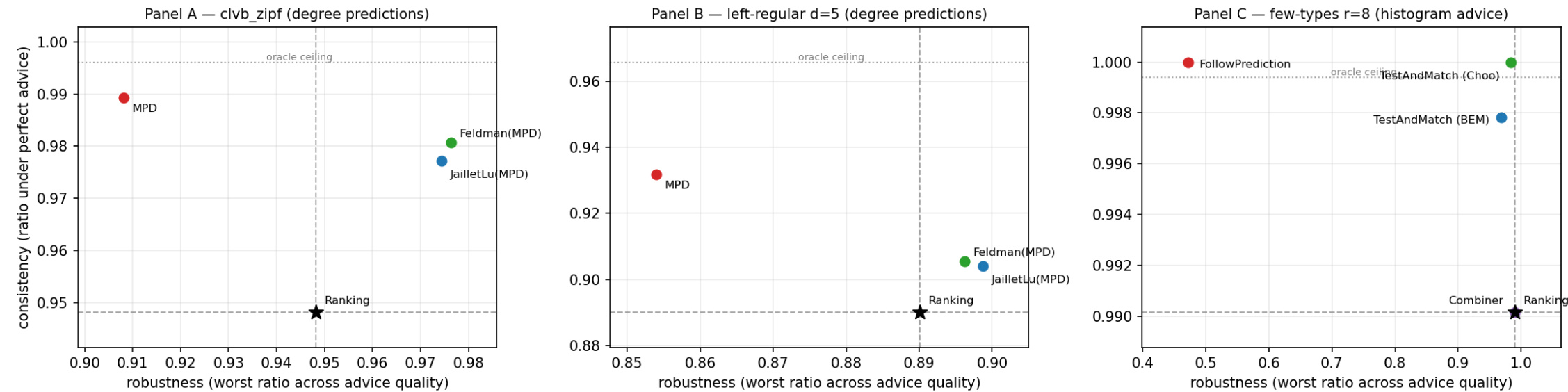
qualitative claims reproduced, all within an absolute difference of 0.02 — despite a different language and a different max-flow routine.

Run on synthetic families, six real-world graphs, and real request traces (Wikipedia, Azure, Mooncake).

Every wide gap is a downside gap

Consistency-robustness plane: up and to the right is better, and the star marks the advice-free floor.

The consistency-robustness plane (data: the unified benchmark, Table 4.1) — up and to the right is better



0.990

advice-free Ranking
already reaches

0.009

total headroom left
for perfect advice

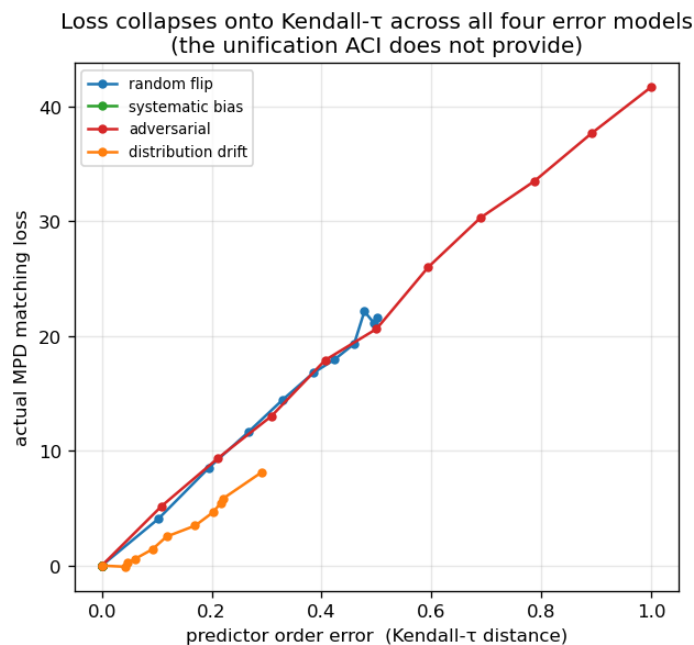
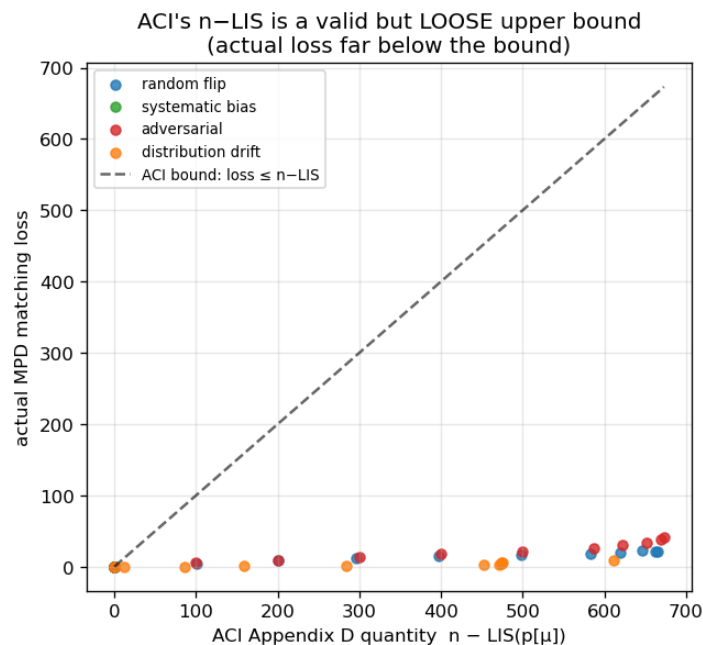
0.472

unguarded prediction-
following collapses to

0.990

the guarded algorithm
holds, whatever the advice

What governs the loss is order, not magnitude



(a) realised loss against the known bound • (b) the same loss against Kendall- τ

- All four error models collapse onto one Kendall- τ curve (Spearman $\rho = 0.979$).
- A monotone rescaling of the predictor changes nothing at all.
- The known $n - LIS$ bound is loose by 16–75 $\times$ and saturates — it cannot tell a harmful prediction from a harmless one.

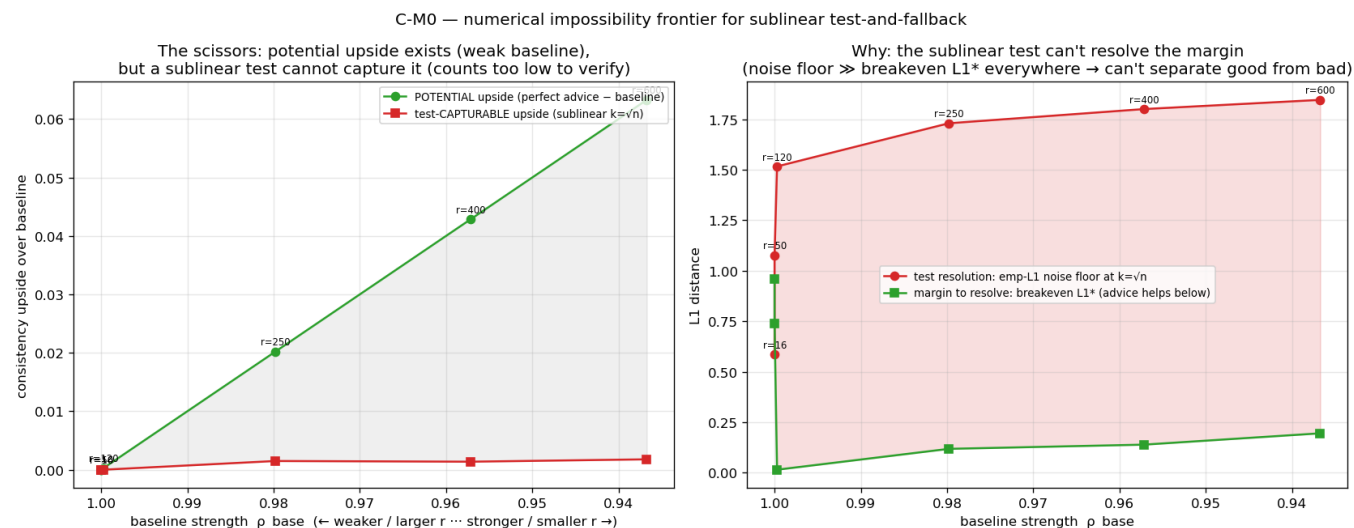
Attribution: order-dependence itself is Aamand-Chen-Indyk's theorem. Mine is the characterisation — which order measure governs the loss, and how loose the known bound is.

A more accurate test can make a worse decision

- The worst-case-calibrated threshold is far too lenient on average-case inputs.
- So a larger, more accurate prefix test decides worse than a small, noisy one.
- Recalibration removes the pathology and exposes what is underneath.

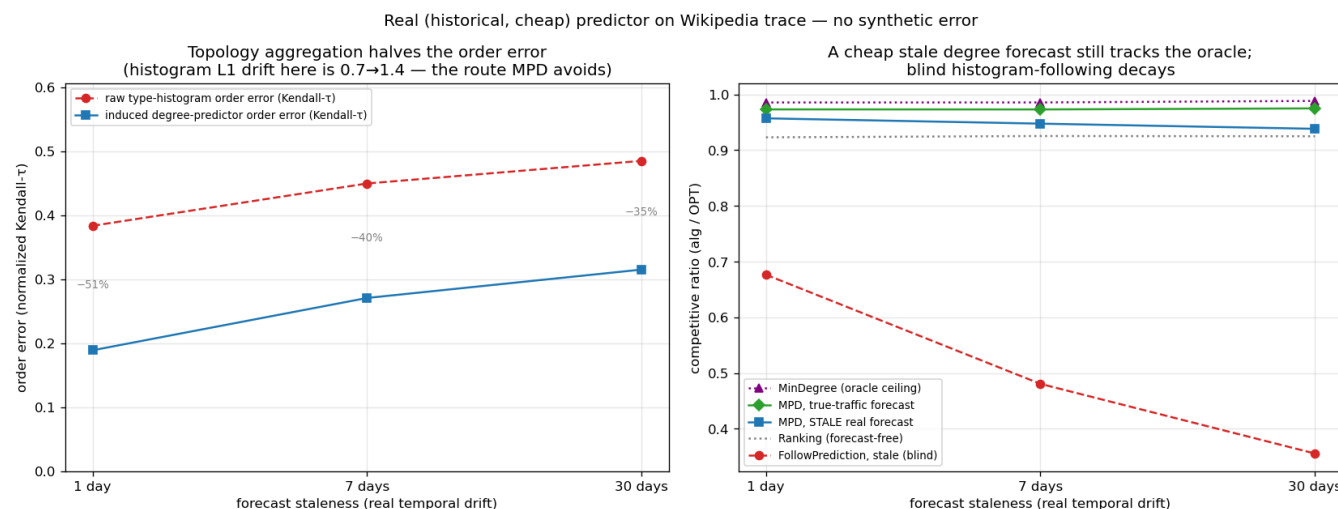
The resolution limit

No practical acceptance threshold can capture an upside smaller than its own estimator's noise — and this holds across the whole difficulty range, not just at one setting.



Potential upside grows as the baseline weakens; the upside a sublinear test can safely capture stays near zero. The curves separate rather than converge.

It survives real predictors and real graphs



A cheap, non-ML predictor: last-window Wikipedia pageview counts, so the error is genuine temporal drift.

Why it works: aggregating the forecast onto the topology halves its order error — and order is all the algorithm consumes.

The same forecast, consumed without a guard, collapses to 0.36 — far below the 0.92 baseline.

27–68%

of the available gap captured by a linear-time count

0.11 ms

to compute, against 4.4 ms for the optimum itself

6 / 6

real-world graphs on which the crash finding still holds

Two attempts to get past the wall

Reported as results, not omitted.

M0

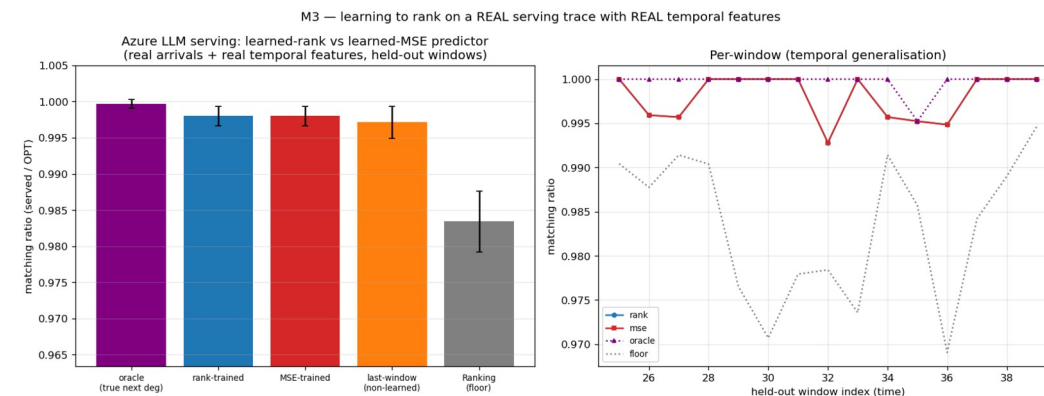
The mechanism is real: with engineered features, training the predictor for order beats training it for accuracy — 0.989 vs 0.974 — while fitting the truth worse.

M1

But the advantage is doubly gated, and peaks at +1.3% of the ratio.

M3

And on genuine temporal features from a real trace it vanishes: identical order ($\tau = 0.126$ both), identical matching.



M3: rank- and MSE-trained predictors are indistinguishable on real features.

Second attempt: a tail objective

If throughput is too forgiving, try latency. Under bursty load the best non-predictive policy comes within 3% of a policy that knows the future exactly.

Why the wall should be expected

Every test-and-fallback algorithm, deciding by any measurable rule on its prefix, satisfies

$$(1 - \eta_c) \leq \eta_r + \gamma_k + o(1)$$

forgone upside ≤ robustness loss + how far apart the two prefix distributions are

- “Consistent and robust” becomes a question of sample complexity: how long a prefix is needed to tell good advice from bad?
- The stakes are capped by the baseline's slack — so on strong-baseline instances the required prefix exceeds the whole instance.

≈ 0.004

the upside the reading says no rule can capture at the benchmark's parameters — exactly the order of the upsides I measured.

Scope: only the inequality above is proved here. The budget law behind the number, and whether it extends past decomposable families, is not established in this thesis.

What the project achieved, and what it did not

1

The brief: met, with one substitution

reproduction and graph types as asked; the extension went to learning-augmented, not streaming — Appendix A.7 now adds that arm

2

My RQ1 and RQ3: met

the theoretical ordering does not match the practical one; the fallback mechanism's real cost is measured in full

3

RQ2: met in part

the order-error curve is stronger than I asked for; where algorithms' curves cross is answered only qualitatively

4

Not delivered: the adversarial arm

my proposal made arrival order an independent variable — I worked only in known-i.i.d., the model where my headline is most likely to hold

With hindsight

Right structured error models — the proposal's own design bet, and i.i.d. noise would have hidden the order effect entirely

Right validating against Borodin et al. before building anything on the harness

Debatable measuring only matching size — a second objective should have run through the whole benchmark

Process I ran the synthetic stages before the real-feature test; the cheaper, decisive experiment should have come first

Where predictions cannot help

1

A cheap, order-faithful predictor already captures nearly all there is to capture.

2

The sophisticated machinery earns its keep as insurance, not as performance.

3

Finding out whether to trust a prediction costs more, on these inputs, than the prediction is worth.

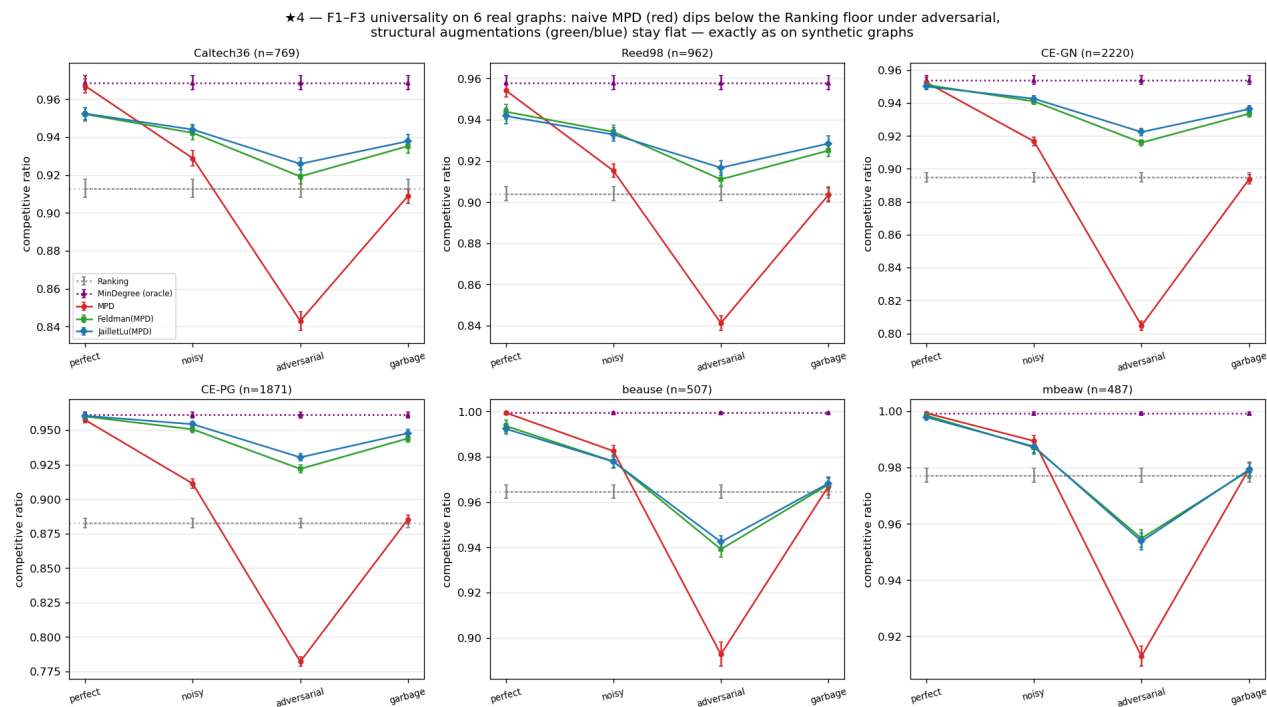
Where to look next

- Adversarial or non-stationary arrivals, where the baseline is provably far from optimal
- Objectives on which the baseline is not near-optimal
- Completing the budget-stakes law

Recognising where predictions cannot help is, I hope, as useful as knowing where they can.

Thank you — happy to take questions.

Backup · the four findings on six real graphs



F1

naive following falls below the advice-free floor on all six, by 0.06 to 0.10

F2

structural robustness holds qualitatively on all six; strictly on the four social and biological graphs

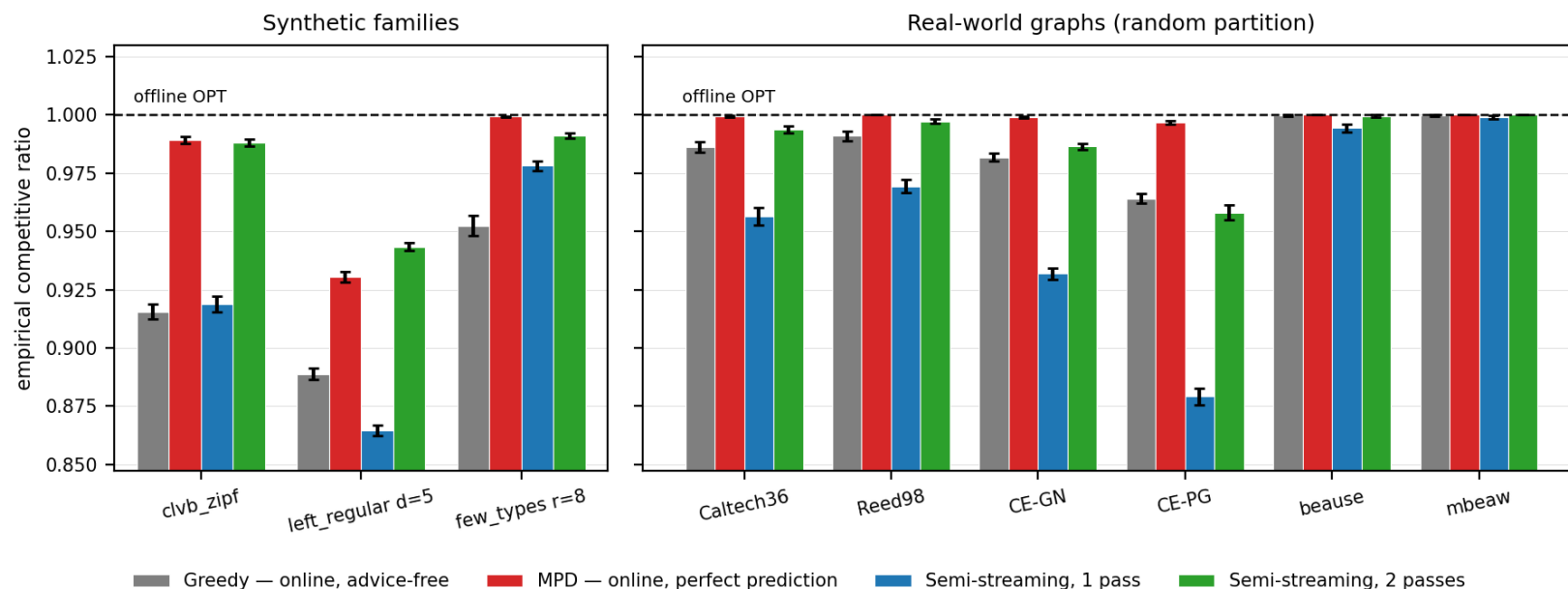
F3

the consistency upside is small everywhere (mean +0.049) and smallest exactly where the baseline is strongest

F4

the augmentation rescues the worst-case-designed algorithms: 0.73–0.77 lifted to 0.99

Backup · the relaxation ladder



The brief suggested streaming or offline algorithms. Placed as a ladder of relaxations of the same instance, they separate the online model's two constraints — bounded memory, and no revision.

One pass is not a free lunch — edge arrival with $O(n)$ memory is worse than online Greedy on seven of nine graphs — 0.879 against 0.964 on CE-PG. Both leave maximal matchings; grouping a node's edges together is what helps.

The second pass is the lever — one revision pass beats a perfect degree prediction on seven of nine, including all six real graphs (CE-PG +0.079 against +0.033).

So the residual gap is irrevocability — not missing information. But a second pass is not an online algorithm — where decisions are physically committed, prediction is the only route.

FOR QUESTIONS

Backup · limitations, stated plainly

Input model

Known-i.i.d. throughout. The wall is an average-case statement; I do not claim it for adversarial arrival order.

Test model

An empirical ℓ_1 surrogate for the unimplemented tolerant tester, following the original authors.

Objective

Matching size only. On tail latency, fairness or recompute cost the picture could differ.

Data breadth

Each real modality is exercised by one trace.

Theory scope

One proved inequality plus a quantitative reading. No theorem beyond that is claimed.

Scale: 16 figures, all regenerated from a fixed seed by a single script; reproduction guide in Appendix A.